

**MIT Art Design and Technology University
MIT School of Computing, Pune**



**Mathematical Foundation for Computing -I
UNIT – I
Sets and Propositions-PART I**



Contents of Unit – I:

(6 Hours)

- **Main Topic-1:** Sets, Combination of sets, Finite and Infinite sets, Uncountably infinite sets, Principle of inclusion and exclusion, Multi-sets, Mathematical Induction.
- **Main Topic-2:** Propositions, Conditional Propositions, Logical Connectivity, Propositional calculus
- **Main Topic-3:** Normal forms, Methods of Proofs, Rules of inference, Predicate calculus, Universal and Existential Quantifiers.

Lecture Sequence



L. No.	Topics
01	Sets, Combination of sets, Finite and Infinite sets
02	Principle of inclusion and exclusion
03	Multi-sets, Mathematical Induction.
04	Propositions, Conditional Propositions, Logical Connectivity
05	Propositional calculus
06	Methods of proofs, Rules of inference, Predicate calculus
07	Universal and Existential Quantifiers

Department of ASH (Mathematics), MIT School of Computing, MIT ADTU, Pune

Sets



- Definition: A collection of well defined objects is called a set.
- Every member of a set is called an Element.
- Sets are denoted by capital letters A, B, C etc.
- Elements are denoted by small letters a, b, c etc.
- One can write down a set in following manners:
- Describe the set in words: e.g. Set of all desks in the class.
- List the elements in the set: e.g. {1, 2, 3, 4}(use curly braces)
- Set builder notation: $Q = \{ p/q \mid p \text{ and } q \text{ are integers and } q \text{ is non zero} \}$
- Remark: elements in a set cannot be repeated.

Examples



- Number sets:

1. Natural numbers $\mathbb{N} = \{1, 2, 3, 4, \dots\}$

2. Integers $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$

3. Rational numbers $\mathbb{Q}$ = Fractions of integers with repeated block of digits after decimal. e.g. 2.345454545

4. Irrational numbers $\mathbb{Q}^*$ = Numbers with decimal expansion having no repeated block of digits after decimal. e.g. π

5. Real numbers $\mathbb{R}$ = Rationals and Irrationals together.

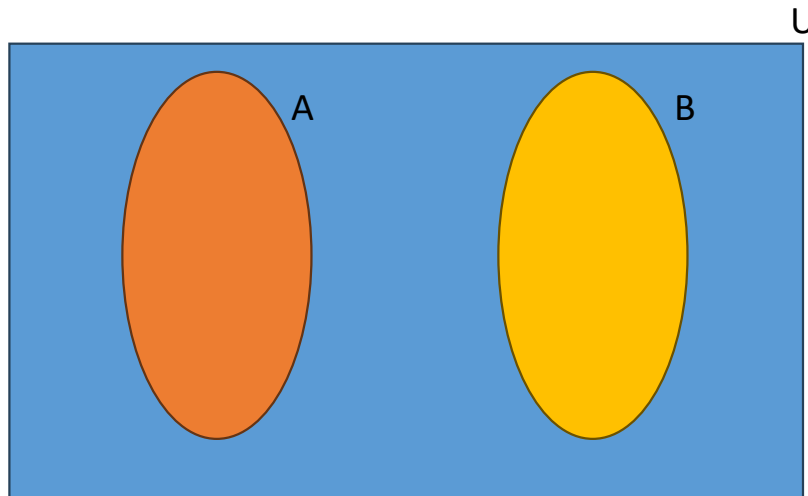
6. Complex numbers $\mathbb{C} = \{a+ib \mid a \text{ and } b \text{ are reals}\}$

Terminology

- **Belongs to:** If x is an element of a set A then we write it as $x \in A$, otherwise $x \notin A$.
e.g. If $A = \{1, 2, 3, 4\}$ then we write $1 \in A$ and $5 \notin A$.
- **Finite set:** A set with finitely many elements is called a finite set. e.g. $A = \{1, 2, 3, 4\}$
- **Infinite set:** A set with infinitely many elements is called infinite set. e.g. $A = \{1, 2, \dots\}$
- **Empty set or Null set ($\emptyset$):** A set with no elements is called empty set. e.g. $A = \{\}$
- **Subset:** If every element of a set B is also an element of set A then we say that B is subset of A , denoted by $B \subseteq A$. e.g. $\mathbb{N} \subset \mathbb{Z}$
- **Equal Sets:** If sets A and B are subsets of each other then they are equal. $A = B$
- **Proper subset:** If B is subset of A but is not the same as A then we say that B is a proper subset of A , denoted by $B \subsetneq A$ or $B \subset A$.
- **Improper subset:** If B is the same as A or B is empty, then we say that B is improper subset of A , denoted by $B = A$ or $B = \emptyset$.
- **Universal set:** If all sets in a context are part of a single set then this single set is called Universal set, denoted by U . e.g. If we talk about rationals and irrationals, we can say that set of reals in U .

Venn Diagrams

- A set can be represented pictorially as follows:



U- rectangle

A, B- ovals

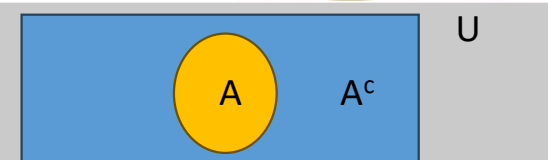
A and B are subsets of U.

- These diagrams help represent finite collection sets and their interactions.

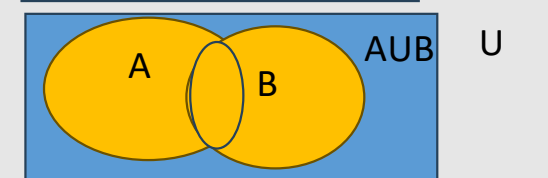
Combination of Sets

Complement of a set: If A is a set then the complement of A is another set with elements in the universal set other than that of A.

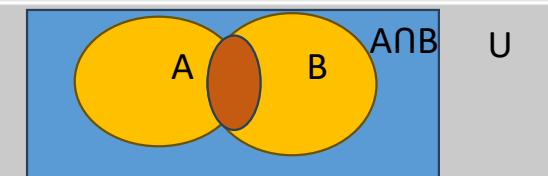
Notation: A^c e.g. $U=\{1,2,3,4\}$, $A=\{1,2\}$, $A^c=\{3,4\}$



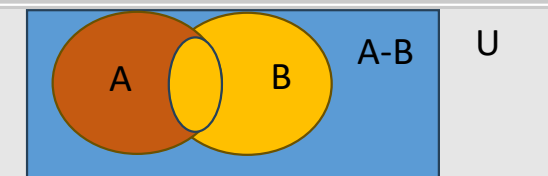
Union of two sets: If A and B are two sets then their union has elements from A or B or both. e.g. $A=\{1,2,3,4\}$, $B=\{1,2,7\}$, $A \cup B = \{1,2,3,4,7\}$



Intersection of two sets: If A and B are two sets then their intersection has common elements from A and B. e.g. $A=\{1,2,3\}$, $B=\{2,3,5,6\}$, $A \cap B = \{2,3\}$

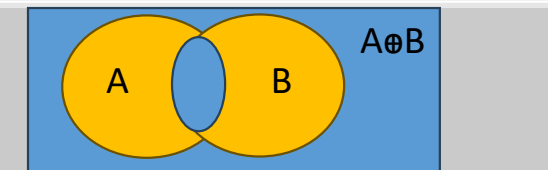


Difference of two sets: If A and B are two sets then their difference $A-B$ or $A \setminus B$ is set of elements of A that are not in B. e.g. $A=\{1,2,3,4,5\}$, $B=\{3,4,6,7,8\}$, $A-B=\{1,2\}$



Symmetric Difference of two sets: If A and B are two sets then their symmetric difference is the set with elements from $A-B$ or $B-A$.

e.g. $A=\{1,2,3,4\}$, $B=\{3,5,6,7\}$, $A \oplus B = \{1,2,5,6,7\}$



More Sets



- **Singleton set**: A set with only one element is called singleton set.
e.g. $A = \{1\}$
- **Doubleton set**: A set with only two elements is called doubleton set.
e.g. $A = \{1, 2\}$
- **Disjoint sets**: If the intersection of two sets A and B is empty set then A and B are said to be disjoint sets.
e.g. $A = \{1, 2, 3, 4\}$, $B = \{5, 6, 7\}$, $A \cap B = \emptyset$
- **Cartesian Product of two sets**: $A \times B = \{(x, y) / x \text{ is in } A \text{ and } y \text{ is in } B\}$
e.g. $A = \{1, 2\}$, $B = \{3, 4, 5\}$, $A \times B = \{(1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5)\}$
- **Power set**: Set of all subsets of any set.
e.g. $A = \{1, 2, 3\}$, $P(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, A\}$

Properties

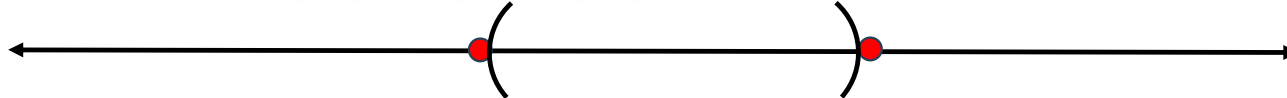
- $A^c = U - A$
- $A - A = \emptyset$
- $A - A^c = A, A^c - A = A^c$
- $A - \emptyset = A$
- $A - B = A \cap B^c$
- $A - B = B - A$ if and only if $A = B$
- $A - B = A$ if and only if $A \cap B = \emptyset$
- $A - B = \emptyset$ if and only if $A \subseteq B$
- $A \oplus A = \emptyset$
- $A \oplus \emptyset = A$
- $A \oplus U = A^c$
- $A \oplus A^c = U$
- $A \oplus B = A \cup B - A \cap B$

Algebra of Set Operations

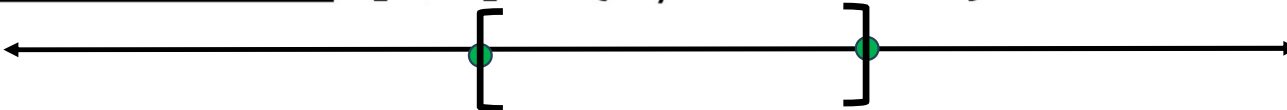
- **Commutativity**: $A \cup B = B \cup A$, $A \cap B = B \cap A$
- **Associativity**: $A \cup (B \cup C) = (A \cup B) \cup C$, $A \cap (B \cap C) = (A \cap B) \cap C$
- **Distributivity**: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$, $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- **Idempotency**: $A \cup A = A$, $A \cap A = A$
- **Absorption Laws**: $A \cup (A \cap B) = A$, $A \cap (A \cup B) = A$
- **De' Morgan's Laws**: $(A \cup B)^c = A^c \cap B^c$, $(A \cap B)^c = A^c \cup B^c$
- **Double Complementation**: $(A^c)^c = A$

Intervals in Real Line

- **Open interval:** $(a, b) = \{x / a < x < b\}$

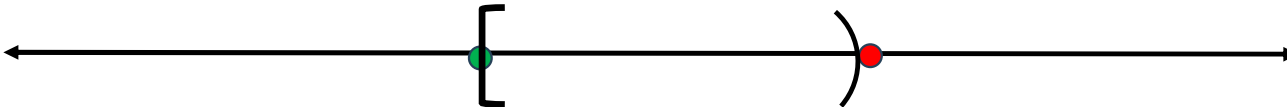


- **Closed interval:** $[a, b] = \{x / a \leq x \leq b\}$



- **Semi open intervals/Semi closed intervals:**

$$(a, b] = \{x / a < x \leq b\}, \quad [a, b) = \{x / a \leq x < b\}$$

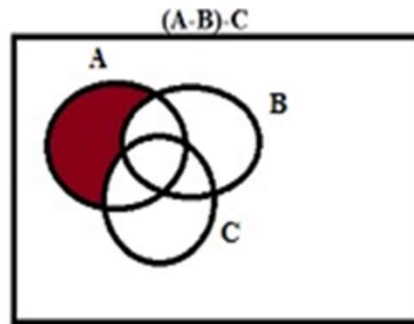
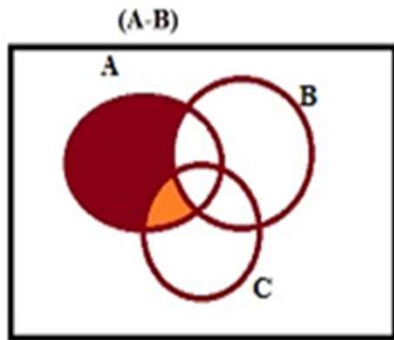


Exercise

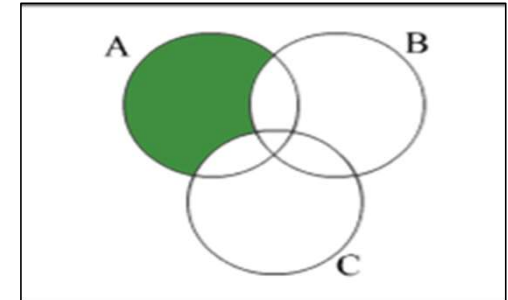
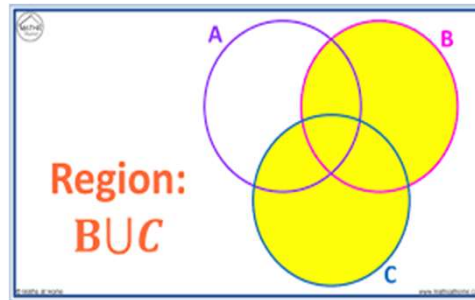
• Show that: $(A - B) - C = A - (B \cup C)$ using Venn diagram.

Solution :

$$(A - B) - C$$



$$A - (B \cup C)$$



Exercise



- Consider the following: a) p : Anil is rich. b) q : Kanchan is poor.

Write each of the following statements in symbolic forms:

- Anil is not rich and Kanchan is poor.
- It is not true that Anil and Kanchan are both rich.
- Either Anil is poor or Kanchan is poor.
- Either Anil or Kanchan is rich.

Exercises

- List the number of sets: $A = \{x \in \mathbb{N} / x < 12\}$
- Use Venn diagram to illustrate the relationships $A \subset B$ and $A \subset C$
- For each of these pairs of sets, determine whether the first is a subset of the second, the second is a subset of the first, or neither is a subset of the other.
 - a) the set of airline flights from New York to New Delhi, the set of nonstop airline flights from New York to New Delhi.
 - b) the set of flying squirrels, the set of living creatures that can fly.
- Prove that $\mathcal{P}(A) \subseteq \mathcal{P}(B)$ if and only if $A \subseteq B$.
- Determine whether each of these statements is true or false.
 - a) $x \in \{x\}$ b) $\{x\} \subseteq \{x\}$ c) $\{x\} \in \{x\}$ d) $\{x\} \in \{\{x\}\}$ e) $\emptyset \subseteq \{x\}$ f) $\emptyset \in \{x\}$
- Find the sets $A - B$, $A \cap B$, $A - (A \cup B)$ for $A = \{a, b, c\}$, $B = \{a, c, d, f\}$

Thank you.



END OF UNIT I - PART I – LECTURE 1

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UNIT – I
Sets and Propositions-PART I**



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Cardinality of a Set

- **Definition:** The number of elements in a set is called the cardinality of a set.
- **Notation:** $|A|$ or $card(A)$
- **Remark:** If $A = \emptyset$ then $|A| = 0$ and for $A \subseteq B$, $|A| \leq |B|$.
- **Countable set:** If a set A is finite or has one-to-one correspondence with the set of natural numbers then we say that the given set is countable or denumerable set, otherwise it called uncountable set.
e.g. $\mathbb{Z}$ – *countable set*, $\mathbb{R}$ – *uncountable set*, intervals of real numbers are uncountable sets.
Ex. Find the $card(P(A))$ where $A = \{1, 2, 3\}$ (observe the pattern in general.)

Principle of Inclusion-Exclusion(PIE)

- **The Addition Principle:** For disjoint and finite sets A and B,

$$|A \cup B| = |A| + |B|$$

- **Theorem:** Let A be a finite set and B be any set(may not be finite), then

$$|A - B| = |A| - |A \cap B|$$

- **PIE statement:** Let A and B be finite sets, then

$$|A \cup B| = |A| + |B| - |A \cap B|$$

- **PIE statement for three sets:**

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$$

Exercise on PIE



- In a discrete mathematics class every student is a major in computer science or mathematics, or both. The number of students having computer science as a major (possibly along with mathematics) is 25; the number of students having mathematics as a major (possibly along with computer science) is 13; and the number of students majoring in both computer science and mathematics is 8. How many students are in this class?

Solution: Let A be the set of students in the class majoring in computer science and B be the set of students in the class majoring in mathematics. Then $A \cap B$ is the set of students in the class who are joint mathematics and computer science majors.

Because every student in the class is majoring in either computer science or mathematics (or both), it follows that the number of students in the class is $|A \cup B|$.

Therefore, $|A \cup B| = |A| + |B| - |A \cap B| = 25 + 13 - 8 = 30$.

Therefore, there are 30 students in the class.

Exercise on PIE

- How many positive integers from first 1000 are divisible by 2 or 3?

Solution: Let A be the set of positive integers from first 1000 that are divisible by 2, and let B be the set of positive integers from first 1000 that are divisible by 3. Then $A \cup B$ is the set of integers from first 1000 that are divisible by either 2 or 3, and $A \cap B$ is the set of integers from first 1000 that are divisible by both 2 and 3. We have

$$|A| = 1000/2 = 500, |B| = 1000/3 = 333.33 \approx 333 \text{ and}$$

$$|A \cap B| = 1000/6 = 166.66 \approx 166$$

It follows that there are $|A \cup B| = |A| + |B| - |A \cap B|$ required number of integers. So, $|A \cup B| = 500 + 333 - 166 = 667$ is the answer.

Exercises on PIE

- Find the number of positive integers not exceeding 100 that are not divisible by 5 or by 7.
- In a survey of 270 college students, it is found that 64 like sprouts, 94 like broccoli, 58 like cauliflower, 26 like both sprouts and broccoli, 28 like both sprouts and cauliflower, 22 like both broccoli and cauliflower, and 14 like all three vegetables. How many of the 270 students do not like any of these vegetables?
- Find the number of elements in $A1 \cup A2 \cup A3$ if there are 100 elements in $A1$, 1000 in $A2$, and 10,000 in $A3$ if
 - a) $A1 \subseteq A2$ and $A2 \subseteq A3$.
 - b) the sets are pairwise disjoint

Exercises on PIE



- In survey of 60 people it was found that: 25 read Business India, 26 read India Today, 26 read Times of India, 11 read both Business India and India Today, 9 read both Business India and Times of India Today, 8 read both India Today and Times of India and 8 read none of these. Find i) How many read at least one? ii) How many read exactly one?

Solution: Let A be the set of all people who read Business India.

Let B be the set of all people who read India Today.

Let C be the set of all people who read Times of India.

So that $|A| = 25$, $|B| = 26$, $|C| = 26$, $|A \cap B| = 11$, $|A \cap C| = 9$, $|B \cap C| = 8$, $|(A \cup B \cup C)^c| = 8$

$$\begin{aligned} \text{i) The no. of people who read at least one} &= |A \cup B \cup C| = 60 - |(A \cup B \cup C)^c| \\ &= 60 - 8 = 52 \end{aligned}$$

$$\begin{aligned} \text{ii) The no. of people who read exactly one} &= |A \cup B \cup C| - |A \cap B| - |A \cap C| - |B \cap C| + 2|A \cap B \cap C| \\ &= 52 - 11 - 9 - 8 + 2|A \cap B \cap C| \end{aligned}$$

Since $|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$, we have $|A \cap B \cap C| = 3$

Hence the answer = 30

Principle of Inclusion-Exclusion(PIE)

Exercise:

Question 1: Consider a set of integers from 1 to 250.

Find how many of these numbers are divisible by 3 or 5 or 7?

Also indicate how many are divisible by 3 or 5 but not by 7.

Question 2: Among the integers 1 to 1000

i)How many of them are not divisible by 3, nor by 5, nor by 7?

ii)How many are not divisible by 5 and 7 but divisible by 3?

Question 3: Among the 130 students, 60 study Mathematics, 51 study Physics and 30 study both Mathematics and Physics. Out of the 54 students studying Chemistry, 26 study Mathematics, 21 study Physics and 12 study both Mathematics and Physics. All the students studying neither Mathematics nor Physics are studying Biology.

- i) How many students are studying Biology? (49)
- ii) How many students not studying Chemistry are studying Mathematics but not Physics?(16)

Question 4: It was found that in the first year of computer science of 80 students, 50 know OOP, 55 know C language and 46 know Python. It was also known that 37 know C and OOP, 28 know C and Python, and 25 know Python and OOP. 7 students however know none of the language. Find

i) How many know all the three language? (12)

ii) How many know exactly two language? (54)

Thank you.



END OF UNIT I - PART I – LECTURE 2

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Multisets(Mset)

- **Definition:** A multiset (short for multiple-membership set) is an unordered collection of elements where an element can occur as a member more than once.

- We can use the same notation for a multiset as we do for a set.

e.g. $A = \{1, 2, 2, 3, 4, 4, 4, 4\}$

- **Multiplicity of an element:** The number of times it occurs in the Mset denoted by μ .

- Each element can be listed the number of times it occurs.

e.g. In Mset A above, $\mu(1)=1$, $\mu(2)=2$, $\mu(3)=1$, $\mu(4)=5$

So $A = \{1 \cdot 1, 2 \cdot 2, 1 \cdot 3, 5 \cdot 4\}$

Terms in Msets

- **Equality**: Elements must be same with same multiplicity.

e.g. $\{a, a, b\} = \{a, b, a\}$ but $\{a, a, b, b\} \neq \{a, a, b\}$

- **Multisubset**: An Mset A is said to be a multisubset of Mset B if all the elements of A are also elements of B with multiplicities of elements of A less than or equal to the multiplicities of the same elements in B.

e.g. $A = \{2 \cdot a, 3 \cdot b\}$ and $B = \{2 \cdot a, 2 \cdot b, 1 \cdot c\}$

- **Union of Msets**: If A and B are two Msets, then $A \cup B$ is the Mset such that for each element $x \in A \cup B$, $\mu(x) = \max(\mu_A(x), \mu_B(x))$ where $\mu_A(x), \mu_B(x)$ are multiplicities of x wrt A and B, resp.

e.g. $A = \{a, b, b, c\}$, $B = \{b, c, c, d\}$, $A \cup B = \{a, b, b, c, c, d\}$

- **Intersection of Msets**: If A and B are two Msets, then $A \cap B$ is the Mset such that for each element $x \in A \cap B$, $\mu(x) = \min(\mu_A(x), \mu_B(x))$ where $\mu_A(x), \mu_B(x)$ are multiplicities of x wrt A and B, resp.

e.g. $A = \{a, b, b, c\}$, $B = \{b, c, c, d\}$, $A \cap B = \{b, c\}$

Terms in Msets

- **Difference of Msets**: If A and B are two Msets, then $A - B$ is the Mset such that for each element $x \in A - B$,

$$\mu(x) = \mu_A(x) - \mu_B(x) , \text{ if difference is greater than zero}$$

$$= 0, \text{ if difference is zero or negative}$$
 where $\mu_A(x), \mu_B(x)$ are multiplicities of x wrt A and B, resp.
 e.g. $A = \{a, b, b, c\}, B = \{b, c, c, d\}, A - B = \{a, b\}$ and $B - A = \{c, d\}$
- **Sum of Msets**: If A and B are two Msets, then $A + B$ is the Mset such that for each element $x \in A + B$, $\mu(x) = \mu_A(x) + \mu_B(x)$ where $\mu_A(x), \mu_B(x)$ are multiplicities of x wrt A and B, resp.
 e.g. $A = \{a, b, b, c\}, B = \{b, c, c, d\}, A + B = \{a, b, b, b, c, c, c, d\}$
- **Symmetric Difference of Msets**: $A \oplus B = (A \cup B) - (A \cap B)$
- **Cardinality of Mset**: Sum of all multiplicities.
 e.g. $A = \{a, b, b, c\}, \text{card}(A) = 4$

Polynomial Representation of Multisets

- A multiset can be represented by a monomial as follows:

1. Empty Mset: $\{ \} \rightarrow x^0 = 1$
2. $\{x\} \rightarrow x^1$
3. $\{x, x\} \rightarrow x^2$
4. $\{x, y\} \rightarrow xy$
5. $\{x, x\} \{y, y\} \rightarrow 2x^2y$
6. $\{x, y\} \{x, y\} \{x, y\} \rightarrow 3xy$
7. $\{x, x, y\} \{x, x, y\} \{x, x, y\} \rightarrow 2x^2y$

Exercises on Msets

- Suppose that P and Q are the multisets $\{4 \cdot a, 1 \cdot b, 3 \cdot c\}$ and $\{3 \cdot a, 4 \cdot b, 2 \cdot d\}$, respectively. Find $P \cup Q$, $P \cap Q$, $P - Q$, and $P + Q$.

Solution: We have

$$\begin{aligned} P \cup Q &= \{\max(4, 3) \cdot a, \max(1, 4) \cdot b, \max(3, 0) \cdot c, \max(0, 2) \cdot d\} \\ &= \{4 \cdot a, 4 \cdot b, 3 \cdot c, 2 \cdot d\} \end{aligned}$$

$$\begin{aligned} P \cap Q &= \{\min(4, 3) \cdot a, \min(1, 4) \cdot b, \min(3, 0) \cdot c, \min(0, 2) \cdot d\} \\ &= \{3 \cdot a, 1 \cdot b, 0 \cdot c, 0 \cdot d\} = \{3 \cdot a, 1 \cdot b\} \end{aligned}$$

$$\begin{aligned} P - Q &= \{\max(4 - 3, 0) \cdot a, \max(1 - 4, 0) \cdot b, \max(3 - 0, 0) \cdot c, \max(0 - 2, 0) \cdot d\} \\ &= \{1 \cdot a, 0 \cdot b, 3 \cdot c, 0 \cdot d\} = \{1 \cdot a, 3 \cdot c\} \end{aligned}$$

$$\begin{aligned} P + Q &= \{(4 + 3) \cdot a, (1 + 4) \cdot b, (3 + 0) \cdot c, (0 + 2) \cdot d\} \\ &= \{7 \cdot a, 5 \cdot b, 3 \cdot c, 2 \cdot d\} \end{aligned}$$

Mathematical Induction

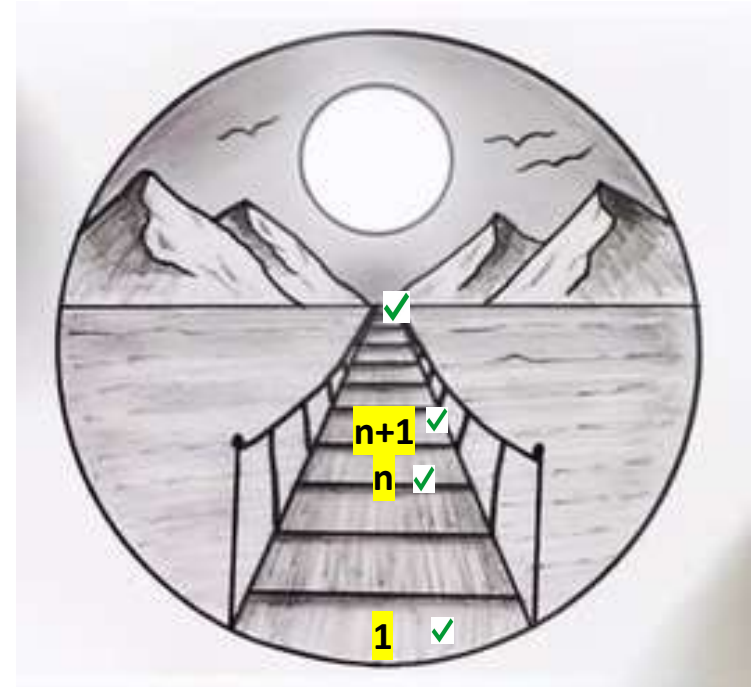
- Climbing a mountain with stairs to watch Sunset.

Suppose we come across a mountain with many stairs.

Suppose we can climb onto the first stair
and can watch the sunset.

Suppose watching the sunset from n th stair
enables us to do the same from the $(n+1)$ th stair,
then the induction tells us that

We can watch the sunset from any stair.



Mathematical Induction



- **Principle of Mathematical Induction(PME)**: Statement

If $P(n)$ is a statement about a natural number n and

- a. if $P(1)$ is true, (base of induction, can start with numbers other than 1 as well)
- b. if $P(k)$ is true implies that $P(k+1)$ is true (induction step, $P(k)$ true is induction hypothesis)

then

$P(n)$ is true for all the natural numbers n .

e.g. the following result $1+2+3+\dots=n(n+1)/2$ can be proved by above principle.

Exercises on PME

- Using Mathematical induction show that: $\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$.

Solution:

Let

$$P(n): \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

$$\text{For } n=1, LHS = \frac{1}{1 \cdot 2} = \frac{1}{2}; RHS = \frac{1}{1+1} = \frac{1}{2}$$

$$\therefore LHS = RHS$$

$$\therefore P(1) \text{ is true.}$$

Let us assume $P(k)$ is true for some $k \in N$

$$\text{i.e., } \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots + \frac{1}{k(k+1)} = \frac{k}{k+1}$$

Adding $(k+1)^{\text{th}}$ term $= \frac{1}{(k+1)(k+2)}$ on both sides,

we get,

$$\begin{aligned} & \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{k(k+1)} + \frac{1}{(k+1)(k+2)} \\ &= \frac{k}{k+1} + \frac{1}{(k+1)(k+2)} \\ &= \frac{k(k+2) + 1}{(k+1)(k+2)} \\ &= \frac{k^2 + 2k + 1}{(k+1)(k+2)} \\ &= \frac{(k+1)^2}{(k+1)(k+2)} = \frac{k+1}{k+2} \\ &= \frac{k+1}{(k+1)+1} \end{aligned}$$

Exercises on PME

- Using Mathematical induction show that $8^n - 3^n$ is divisible by 5 for $n \geq 1$

Solution: Let $P(n)$ be the statement $8^n - 3^n$ is divisible by 5 for $n \geq 1$.

i) Basis of Induction: $P(1)$ is true: $8^1 - 3^1 = 5$ is divisible by 5.

ii) Suppose $8^k - 3^k$ is divisible by 5.

$$\begin{aligned}\text{Consider } 8^{k+1} - 3^{k+1} &= 8 * 8^k - 3 * 3^k = 5 * 8^k + 3 * 8^k - 3 * 3^k \\ &= 5 * 8^k + 3(8^k - 3^k)\end{aligned}$$

Which is divisible by 5 as both terms are divisible by 5.

Hence by the PIE, it is true for all the natural numbers n .

Exercises on PME

- Prove by mathematical induction:

$$1 \cdot 2 + 2 \cdot 3 + \cdots + n \cdot (n + 1) = \frac{n(n+1)(n+2)}{3}$$

- Use mathematical induction to show that

$$1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$$

- Use mathematical induction to prove the inequality $n < 2^n$ for all positive integers n .
- Use mathematical induction to prove that $n^3 - n$ is divisible by 3 whenever n is a positive integer.

Thank you.



END OF UNIT I - PART I – Lecture 3

**MIT Art Design and Technology University
MIT School of Computing, Pune**



Mathematical Foundation for Computing -I
UNIT – I
Sets and Propositions-PART II



Contents of Unit – I:

(6 Hours)

- **Main Topic-1:** Sets, Combination of sets, Finite and Infinite sets, Uncountably infinite sets, Principle of inclusion and exclusion, Multi-sets, Mathematical Induction.
- **Main Topic-2:** Propositions, Conditional Propositions, Logical Connectivity, Propositional calculus
- **Main Topic-3:** Normal forms, Methods of Proofs, Rules of inference, Predicate calculus, Universal and Existential Quantifiers.

Lecture Sequence



L. No.	Topics
01	Sets, Combination of sets, Finite and Infinite sets
02	Principle of inclusion and exclusion
03	Multi-sets, Mathematical Induction.
04	Propositions, Conditional Propositions, Logical Connectivity
05	Propositional calculus
06	Methods of proofs, Rules of inference, Predicate calculus
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Propositions



- **Proposition**: It is declarative sentence/statement which is either true or false, but not both.

e.g. $1+1=2 \rightarrow \text{true}$, earth is flat $\rightarrow \text{false}$

- **Notations**: small letters like p, q, r etc.(called statement variables)
- Can the answer to the following question be a proposition? Think.

The barber is a man in town who shaves all those, and only those, men in town who do not shave themselves.

Does the barber shave himself?

Logical Connectives

Symbol	Connective	Name	Propositions are combined by connectives as follows(statement forms)	Interpretation in terms of logical constants TRUE(T) and FALSE(F) (truth values of logical constants)
$\wedge$	and	conjunction	$p \wedge q$	Is T, if both p and q are T
$\vee$	or	disjunction	$p \vee q$	Is T, if any one of p and q is T
$\Rightarrow$	implies	implication / conditional	$p \longrightarrow q$	If p is T then Q is T (alternatively denoted by $\longrightarrow$)
$\Leftrightarrow$	is equivalent	biconditional	$p \longleftrightarrow q$	p is T if and only if q is T (alternatively denoted by $\longleftrightarrow$)
$\neg$	not	negation	$\neg p$	Is T, if p is F.
$\bar{\vee}$	ex-or	Exclusive or	$p \bar{\vee} q$	Is T, if only one of the p and q is T

Truth Tables

P	$\sim P$
T	F
F	T

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

P	Q	$P \leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Exercises

(i) Truth table for $\neg p \wedge \neg q$

p	q	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	F	F	F
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

(ii) Truth table for $\neg(\neg p \wedge \neg q)$

p	q	$\neg q$	$p \wedge \neg q$	$\neg(\neg p \wedge \neg q)$
T	T	F	F	T
T	F	T	T	F
F	T	F	F	T
F	F	T	F	T

Exercises

(iii) Truth table for $(p \vee q) \vee \neg q$

p	q	$\neg q$	$p \vee q$	$(p \vee q) \vee \neg q$
T	T	F	T	T
T	F	T	T	T
F	T	F	T	T
F	F	T	F	T

(iv) Truth table for $(\neg p \rightarrow r) \wedge (p \leftrightarrow q)$

p	q	r	$\neg p$	$(\neg p \rightarrow r)$	$p \leftrightarrow q$	$(\neg p \rightarrow r) \wedge (p \leftrightarrow q)$
T	T	T	F	T	T	T
T	T	F	F	T	T	T
T	F	T	F	T	F	F
T	F	F	F	T	F	F
F	T	T	T	T	F	F
F	T	F	T	F	F	F
F	F	T	T	T	T	T
F	F	F	T	F	T	F

Exercises



- Write the following sentences in symbolic form:

1. It is very hot and humid, therefore it will rain.

Solution: It is hot : p

It is humid : q

It will rain : r

Symbolic form : $(p \wedge q) \rightarrow r$

2. I am tall but I am not tall.

Solution: I am tall : p

I am not tall : $\neg p$

Symbolic form : $p \wedge \neg p$

Thank you.



END OF UNIT I - PART II – Lecture 1

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MIT School of Computing, Pune**



Mathematical Foundation for Computing -I
UNIT – I
Sets and Propositions-PART II



Contents of Unit – I:

(6 Hours)

- **Main Topic-1:** Sets, Combination of sets, Finite and Infinite sets, Uncountably infinite sets, Principle of inclusion and exclusion, Multi-sets, Mathematical Induction.
- **Main Topic-2:** Propositions, Conditional Propositions, Logical Connectivity, Propositional calculus
- **Main Topic-3:** Normal forms, Methods of Proofs, Rules of inference, Predicate calculus, Universal and Existential Quantifiers.

Lecture Sequence



L. No.	Topics
01	Sets, Combination of sets, Finite and Infinite sets
02	Principle of inclusion and exclusion
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Terminology in Propositional Calculus

- **Tautology**: A logical statement which is always true irrespective of the truth values of the individual statement variables.

e.g. $p \vee \neg p$ (check with the truth table.)

- **Contradiction**: A logical statement which is always false irrespective of the truth values of the individual statement variables.

e.g. $p \wedge \neg p$ (check with the truth table.)

- **Contingency(Neither)**: A logical statement which is neither tautology nor contradiction.

e.g. $p \rightarrow q$

Example 01:

Consider the following:

a) p : Anil is rich. b) q : Kanchan is poor.

Write each of the following statements in symbolic forms:

i) Anil is not rich and Kanchan is poor.

ii) It is not true that Anil and Kanchan are both rich.

iii) Either Anil is poor or Kanchan is poor.

iv) Either Anil or Kanchan is rich.

Solution:

p: Anil is rich

q: Kanchan is poor

i) Anil is not rich and Kanchan is poor. Symbolic form: $\neg p \wedge q$

ii) It is not true that Anil and Kanchan are both rich. Symbolic form: $\neg(p \wedge \neg q)$

iii) Either Anil is poor or Kanchan is poor. Symbolic form: $\neg p \vee q$

iv) Either Anil or Kanchan is rich. Symbolic form: $p \vee \neg q$

Example 02:

Consider the following: a) p : Gopal is intelligent. b) q : Gopal is rich. Write the logical negation of the following statements in symbolic forms:

- i) Gopal is intelligent and rich.
- ii) Gopal is intelligent but not rich.
- iii) Gopal is either intelligent or rich.

Solution:

Given Statements:

p: Gopal is intelligent

q: Gopal is rich

i) Gopal is intelligent and rich.

Given statement: $p \wedge q$

Negation: $\neg(p \wedge q)$

The negation of "Gopal is intelligent and rich" is "It is not the case that Gopal is intelligent and rich." Symbolically, this is represented as $\neg(p \wedge q)$.

ii) Gopal is intelligent but not rich.

- Given statement: $p \wedge \neg q$
- Negation: $\neg(p \wedge \neg q)$

The negation of "Gopal is intelligent but not rich" is "It is not the case that Gopal is intelligent but not rich." Symbolically, this is represented as $\neg(p \wedge \neg q)$.

iii) Gopal is either intelligent or rich.

- Given statement: $p \vee q$
- Negation: $\neg(p \vee q)$

The negation of "Gopal is either intelligent or rich" is "It is not the case that Gopal is either intelligent or rich." Symbolically, this is represented as $\neg(p \vee q)$.

Example 03:

Consider the following: a) p : This system is good. b) q : This system is cheap.

Write each of the following statements in symbolic forms:

- i) This system is good and cheap.
- ii) This system is not good but cheap.
- iii) This system is neither good nor cheap.
- iv) This system is good or cheap.

Solution:

Given Statements:

p: This system is good

q: This system is cheap

i) This system is good and cheap.

Symbolic form: $p \wedge q$

ii) This system is not good but cheap.

- Symbolic form: $\neg p \wedge q$

In this statement, "not good" is represented by $\neg p$, and "cheap" is represented by q . Both conditions are connected using the conjunction operator $\wedge$, resulting in the symbolic form $\neg p \wedge q$.

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iii) This system is neither good nor cheap.

Symbolic form: $\neg p \wedge \neg q$

This statement implies that the system is neither good ($\neg p$) nor cheap ($\neg q$).

Both conditions are connected using the conjunction operator $\wedge$, resulting in the symbolic form $\neg p \wedge \neg q$.

iv) This system is good or cheap.

Symbolic form: $p \vee q$

Exercises



- Determine which of the following is tautology, contradiction, neither?
 1. $p \rightarrow (q \rightarrow r)$
 2. $(p \vee q) \wedge (\neg p \vee \neg q)$
 3. $(\neg p \vee q) \longleftrightarrow (p \rightarrow q)$
 4. $\neg p \longleftrightarrow p$
- Write the statement forms and check for tautology:
 1. I am tall therefore I am short and I am good.
 2. I do not study so I do pass and do study.

Terminology



- **Equivalence of Statement Forms:**

Two statement forms $S1$ and $S2$ are said to be equivalent if both have the same truth values, irrespective of the truth values of the individual statement variables.

e.g. $p \Leftrightarrow (p \wedge p)$, $(p \rightarrow q) \Leftrightarrow (\neg q \rightarrow \neg p)$

Observe: $S1 \longleftrightarrow S2$ is tautology.

- **Logical Implication:**

A statement form $S1$ is said to logically imply another statement form $S2$ if $S1 \rightarrow S2$ is tautology.

e.g. $(p \wedge q) \Rightarrow (p \rightarrow q)$

Exercises



- Show that $p \rightarrow (q \rightarrow r)$ and $p \rightarrow (\sim q \vee r)$ are logically equivalent using truth table.
- Show that $p \rightarrow (q \rightarrow r)$ and $((p \rightarrow q) \rightarrow (p \rightarrow r))$ are logically equivalent using truth table.
- Show that $p \leftrightarrow q$ and $(p \wedge q) \vee (\sim p \wedge \sim q)$ are logically equivalent using truth table.

Normal Forms



- A logical statement may have a large number of variables or connectives.
- e.g. $[(p \vee q) \wedge (p \leftrightarrow \neg p)] \rightarrow (r \rightarrow p)$
- Drawing conclusion about the statement may not be easy thus.
 - Drawing of truth tables may not be feasible as well.
 - Such statements can be brought down to some simple form.
 - We call it Normal Form.
 - There are different types of normal forms.(canonical forms)

Disjunctive Normal Form(DNF)



- Statement form consisting of disjunction of conjunctions.
- Also said to be OR of ANDs.
- Looks like: $(\wedge) \vee (\wedge) \vee (\wedge) \vee \dots$

e.g. $(p \wedge p) \vee (q \vee r)$

Ex. Reduce the following to DNF. (Use $\equiv$ symbol for equivalence)

$$(p \rightarrow q) \wedge (\neg p \wedge q)$$

$$\begin{aligned}\text{Solution: } (p \rightarrow q) \wedge (\neg p \wedge q) &\equiv (\neg p \vee q) \wedge (\neg p \wedge q) \\ &\equiv (\neg p \wedge \neg p \wedge q) \vee (q \wedge \neg p \wedge q) \\ &\equiv (\neg p \wedge q) \vee (q \wedge \neg p)\end{aligned}$$

Conjunctive Normal Form(CNF)



- Statement form consisting of conjunction of disjunctions.
- Also said to be AND of ORs.
- Looks like: $(\vee) \wedge (\vee) \wedge (\vee) \wedge \dots$

e.g. $(p \vee p) \wedge (q \vee r)$

Ex. Reduce the following to DNF. (Use $\equiv$ symbol for equivalence)

$$(p \rightarrow q) \wedge (\neg p \vee q)$$

$$\begin{aligned}\text{Solution: } (p \rightarrow q) \wedge (\neg p \vee q) &\equiv (\neg p \vee q) \wedge (\neg p \vee q) \\ &\equiv (\neg p \vee q)\end{aligned}$$

Exercises



- Find the CNF and/or DNF of the following:

1. $p \rightarrow (q \rightarrow r)$

2. $(p \rightarrow q) \rightarrow r$

3. $(p \rightarrow q) \vee (q \rightarrow p)$

4. $(\neg p \vee q) \rightarrow r$

5. $[(p \rightarrow r) \rightarrow (q \rightarrow p)] \rightarrow q$

Thank you.



END OF UNIT I - PART II – Lecture 2

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Mathematical Foundation for Computing -I
UNIT – I
Sets and Propositions-PART III



Contents of Unit – I:

(6 Hours)

- **Main Topic-1:** Sets, Combination of sets, Finite and Infinite sets, Uncountably infinite sets, Principle of inclusion and exclusion, multi-sets, Mathematical Induction.
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Lecture Sequence



L. No.	Topics
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Quantifiers

- Expression that asserts the existence of a property of phenomenon inside a domain somewhere or everywhere.
- Basically it talks about the bulk or quantity of entities satisfying a particular property.

e.g. All of the students of the course MFC-I for the domain of FYBTech CSE-IT.

Here entities are Students, property is enrolled for MFC-I, quantity is class FYBTech CSE-IT student strength.

- **Universe of Discourse**: The set or domain from where the entities/values are chosen.

Universal and Existential Quantifiers

- **Universal Quantifier**: For all values of x , the assertion $p(x)$ is true or false, are statements in which we say that the variable x is universally quantified.

- **Notation**: $\forall x, p(x) \text{ is true}$ or $\forall x, p(x) \text{ is false}$

e.g. all the even integers are divisible by 2. $\forall n, n \% 2 = 0, n \in \mathbb{Z}$

- **Existential Quantifier**: Suppose the assertion $p(x)$ is true doesn't hold for one particular value of x , then we say that there exists x such that $p(x)$ is false.

e.g. all the even integers are divisible by 4. $\exists n, n \% 2 \neq 0, n \in \mathbb{Z}$ (for $n=2$)

Negation of Quantification



- $\forall x, p(x) \Leftrightarrow \exists x, \neg p(x)$
- So Universal Quantifier gets negated by the Existential Quantifier and vice versa.
- For statements with more than one variable, the negation may work differently.

e.g. $\forall x \forall y, p(x, y)$ gets negated to $\forall x, \exists y, \neg p(x, y)$

e.g. A pair of a brother and sister is called good, if each of them is good.
Consider the statement, “all the pairs of brothers and sisters are good.”

What will be the negation of the statement?

Exercises

- Translate the following into symbolic form:
 1. Square of every natural number is positive.
 2. There are few students who rank 1 in the examinations.
 3. All the birds can fly.
- Negate the following:
 1. $\forall y, y \neq 2$
 2. $\exists z, z^3 < 1000$
- Write in symbolic form:
 1. All the FYBTech students know MFC-I. Ram is a FYBTech students. So Ram knows MFC-I.
 2. All the teachers teach good. Sham does not teach good. So Sham is not a teacher.

Thank you.



END OF UNIT I - PART III – Lecture 2 - UNIT ends here.